PALEOMAGNETISM OF THE MIDDLE MISSISSIPPIAN GREENBRIER GROUP IN WEST VIRGINIA, USA

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Abstract. The paleomagnetism of the Middle Mississippian Greenbrier Group along the boundary between the Appalachian Plateau province and the Valley and Ridge province has been studied with broad coverage throughout its outcrop area in West Virginia. Multicomponent analysis based on examination of Zijderveld diagrams for both AF and thermal demagnetization results reveals two directions characteristic of the unit. One is the present-day field; the other yields a characteristic direction with southeasterly clustering. Rocks of different magnetic behavior may be distinguished by their lithofacies: gray sediments and red sediments. Demagnetization ranges of 20-30 mT and a heating temperature of 300°C were required to isolate the main magnetic component of gray sediments. Thermal cleaning is more effective in recovering the characteristic direction. Principal component analysis has been applied to determine the best-fitting straight-line segment of the characteristic component during demagnetization and also to test data for non-circular distribution about the mean direction. This test showed no evidence of significant contamination by the recent magnetic field, but both gray and red sediments failed the fold test at the three southernmost sampling sites. Comparison of unfolded VGPs indicates a possible remagnetization time to be Late Carboniferous to Early Permian for red sediments. The presence of both normal and reversed polarities in some gray sediments suggests that remagnetization may be restricted to the overturned beds on the Virginia-West Virginia border. If this is the case, then a pole position based only on thermally demagnetized gray sediments from relatively flat-lying beds may be determined at S 43.2° W 49.5° with k = 62 and dP = 4° and dM = 8°. Because part of the Greenbrier Group is clearly remagnetized, and because a meaningful fold test is not available for the sites used in determining this pole, its reliability is only poor to fair. It agrees quite well, however, with other Mississippian poles for cratonic North America. An analysis of Mississippian poles from cratonic North America and the Acadian displaced terrane indicates that Acadia was some 14° south of its present position and may have increased significantly its northward velocity during the Mississippian.

Introduction

A paleomagnetic study of the Middle Mississippian Greenbrier Group has been undertaken with a view to investigate the character of the geomagnetic field during Carboniferous time and to understand better the value and role played by Paleozoic limestones as magnetic recorders. Previous studies of limestones [Graham, 1956; McAulay and Morris, 1971; Morris, 1971; McElhinney and Opdyke, 1973; Martin, 1975; Kent, 1979] suggest that they often yield valuable paleomagnetic data.

Irving [1979] points out that there are few reliable paleomagnetic poles for cratonic North America in the Carboniferous. A recent summary of available poles has been given by Barton and Brown [1983]. Since speculations on displaced terranes such as the Acadian [Kent and Opdyke, 1978; Van der Voo, 1982] are based on comparison with cratonic poles, it is of considerable importance to establish the reference poles to as great a precision as possible. Carboniferous sediments are abundant in the Appalachian Basin, and many lithofacies have been found to yield characteristic magnetization directions which are consistent with those reported from other areas [Cain et al., 1979].

Several paleomagnetic pole positions resulting from previous work in the Appalachian Basin have been reported [Payne et al., 1981; Shulik, 1979; Noltimier and Kopacz, 1975] covering the Pennsylvanian and Upper Mississippian, and the magnetic remanence was assumed to align along the Carboniferous geomagnetic field at or shortly after deposition. These sediments are nearly flat-lying, however, and so are not amenable to Graham's [1949] fold test. Thus there is no real proof that the magnetization of these units is not diagenetic and related to geological events

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SYSTEM	SERIES	SOUTHEAST		NORTH
		GROUP	FORMATION	FORMATION
MISSISSIPPIAN	Chesterian	Mauch Chunk	Bluestone Formation[1]	Mauch Chunk Formation
			Princeton Sandstone	
			Hinton Formation	?
			Bluefield Formation	Greenbrier Limestone
		Greenbrier	Alderson Limestone	
			Greenville Shale	
			Union Limestone	
			Pickaway Limestone	
			Taggard Formation	
	Meramecian		Denmar Formation	Pocono Formation
			Hillsdale Limestone	
	Osagean	Maccrady Formation[2]		
	Kinderhookian	Pocono Formation		Hampshire Formation[3]

[1]In extreme southern West Virginia, uppermost beds of Bluestone Formation are Pennsylvanian

[2]Lower Maccrady Formation may be Osagean

[3]Most of Hampshire Formation is Devonian

Fig. 1. Generalized section of Greenbrier Group in West Virginia [Arkle et al., 1979].

that might significantly postdate the deposition of the sediments.

The Middle Mississippian Greenbrier Group with its steeply-dipping outcrops near the Virginia border in south-eastern West Virginia was chosen as a means of approaching this problem.

Geology and Sampling

The Greenbrier Group of West Virginia is present in part of the Appalachian Plateau province and part of the synclinal-mountain area of the Valley and Ridge province farther east [Arkle et al., 1979]. The group, deposited in a shallow sea, consists overwhelmingly of limestones, interbedded with calcareous shales, siltstones, and sandstones [Leonard, 1968].

Exposed sections of the Greenbrier reach a maximum thickness of 600 m in Mercer County, at the extreme south in West Virginia. It thins to the north, and in Randolph County the Greenbrier Group changes drastically with the lower formations pinching out. To the south, the limestones generally include more micrite and oolite, whereas the northern Greenbrier contains more clastics. Greenbrier carbonates are classified by differing proportions of matrix micrite, coarser calcitic grains and dolomite. In the geologic reports of Mercer, Monroe, and Summers Counties, Reger and Price [1926] divided the thick Greenbrier section exposed there into 11 members, while Arkle et al. [1979] reduce these to seven (Figure 1). Among these the Taggard Formation, a three-fold series of red shale, limestone (thin, variable, usually well-weathered), and red shale is a reliable correlation marker.

The area investigated in this study (Figure 2) is confined to Mercer, Monroe, Greenbrier, Pocahontas and Randolph Counties. In Mercer County, the Greenbrier crops out in a narrow belt of limestone which roughly parallels the Valley and Ridge Province directly to the east. The Greenbrier there is overturned and dips to the southeast. The beds north of central Monroe County gradually assume a horizontal position, and north of Union and continuing to Edray in Pocahontas County, the beds are essentially horizontal [McCue et al., 1939]. At the western limits of the outcrop, the Williamsburg and Elkins anticlines provide dips up to 30°, but they are unfortunately oriented with respect to the remanence direction for a fold test.

The stratigraphy of the Greenbrier sediments has been studied by Leonard [1968], and his work has been used in locating sites suitable for paleomagnetic study. However, during field work, similarities of the lithologic features of the limestones made division into separate members very difficult, so often the individual limestone members could not be recognized with certainty. Appearance of the Taggard member in some of the outcrops was a good marker in placing rocks within the upper or lower Greenbrier Group.

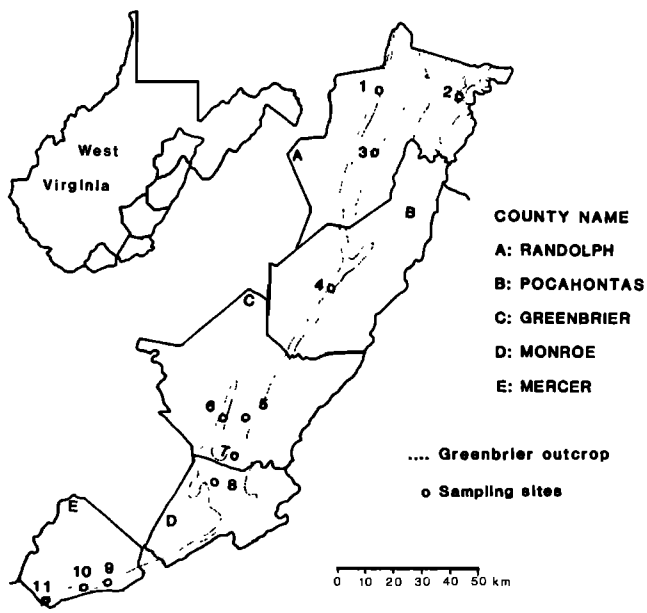


Fig. 2. Index map showing area of study. Sampling sites shown by circles, Greenbrier surface exposures outlined by dotted lines. Site descriptions are given in the Appendix.

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The results fall into two groups distinguished by their lithofacies. These are gray sediments: limestones with varied shades of gray color and rock components (oolitic , micritic or clastic); and red sediments: the reddish shales (sandy or calcareous).

Paleomagnetic Measurements

Measurements of the natural remanent magnetization (NRM) were carried out with a large-bore ScT cryogenic magnetometer on specimens cut to cylindrical shape 3.45 cm in diameter and 2.8-3.5 cm high. Two or six orientations were measured for each specimen, the latter used for weak specimens (total moment less than 5×10^{-8} A m^2/Kg) which results in a total of 18 measurements for our three axis system. Consistent magnetizations from samples with demagnetized intensity as low as 10^{-8} A m^2/Kg can still be obtained. Sample NRM intensities ranged between 2×10^{-7} to 7×10^{-6} A m^2/Kg for red sediments, and 2×10^{-8} to 1×10^{-5} A m^2/Kg for gray sediments, higher values typically associated with the Taggard red member, and lower values with gray limestone members.

A total of 141 gray limestones and 32 red sediment samples were collected from 11 sites (see Appendix for site descriptions). Most samples were collected as tripod-oriented hand samples. Up to six specimens were drilled in the laboratory from each sample, depending on the size of the sample. Twenty-six of the gray samples (collected in the preliminary stages of the study) received only stepwise alternating field (AF) demagnetization. Two gray and 14 red samples received only thermal demagnetization, while 113 gray and 18 red samples received both AF and thermal treatment. Demagnetization diagrams are plotted in Figure 3 according to the Zijderveld method [Zijderveld, 1967].

Gray Sediments

Different types of behavior were found during AF cleaning of gray sediments. In addition to the few weak and viscous samples which showed a random changing of direction (Figure 3a), a large part of the samples showed a steady removal of magnetization while others showed a prominent shift to a fairly stable direction in the demagnetization range of 20-30 mT (Figure 3b). This produces what is referred to as a "streak" between the steep directions identified with the recent field and the characteristic directions with southeasterly declinations and shallow inclinations (Figure 4).

Many limestone samples were dominated by recent secondary magnetizations which could not be removed adequately by AF demagnetization (Figure 3c) before the limit of the demagnetizer was reached. However, many of these revealed a southeasterly clustering during thermal treatment above about 300°C (Figures 3d and 3e), and the directions are well-grouped (Figure 4a) for further higher demagnetization temperatures until finally the intensities either fall below the noise-dominated level by 500°C or increase because of heat-induced alteration of the magnetic minerals.

It is thus reasonable to assume that the vector being removed up to 300°C for thermal demagnetization and 20-30 mT (200-300 oe) for AF demagnetization is probably a viscous remanence showing overprinting by the present geomagnetic field. The final trajectory, on the basis of consistency of directions from sample to sample, can be considered characteristic of these rocks.

Curvature in the demagnetization trajectories to somewhat higher demagnetization fields on some samples (Figure 3b) is interpreted as due to a broader, although not complete, overlap in the coercivity spectra of secondary and characteristic components.

Magnetizations were reduced typically to below 40% of the NRM after being heated to the temperature range of 200°-300°C. A slow thermal decay was associated typically with weak intensity samples, on the order of 10^{-8} A m^2/Kg (Figure 5 Curve A). An alternating field of less than 30 mT was required to reduce sample intensities to half of their original values (Figure 6).

Isothermal remanence (IRM) acquisition (Figure 7) for gray sediments showed the greatest activity at fields below 100 mT and an approach to saturation by 500 mT.

Taken together, the magnetic behavior of the gray sediment indicates that the principal carriers of remanence are fine-grained magnetite or titanomagnetite with some contribution in the very high coercivity range due to hematite.

Red Sediments

Most of the red sediments did not respond well to AF cleaning (Figure 3f) while thermal treatment showed stable directions of magnetization with southeasterly clustering and no significant change in direction from room temperature to 650°C (Figure 4c).

The red sediments responded sluggishly to IRM acquisition (Figure 7), showing little activity below 50 mT and no sign of saturation by 1000 mT.

Normalized thermal decay of NRM curves (Figure 8) exhibit wide variations. Except for only a few samples, most of the remanence in these specimens resides in grains having critical blocking temperatures of less than 600°C. The IRM acquisition and thermal demagnetization studies thus indicate that the magnetic mineral present is probably very fine-grained hematite. A few curves in Figure 8 display a distinct inflection below 600°C and may reflect the presence of magnetite as well.

Optical identification of the magnetic minerals for both groups is difficult, for in polished sections the grain-size was too small and the

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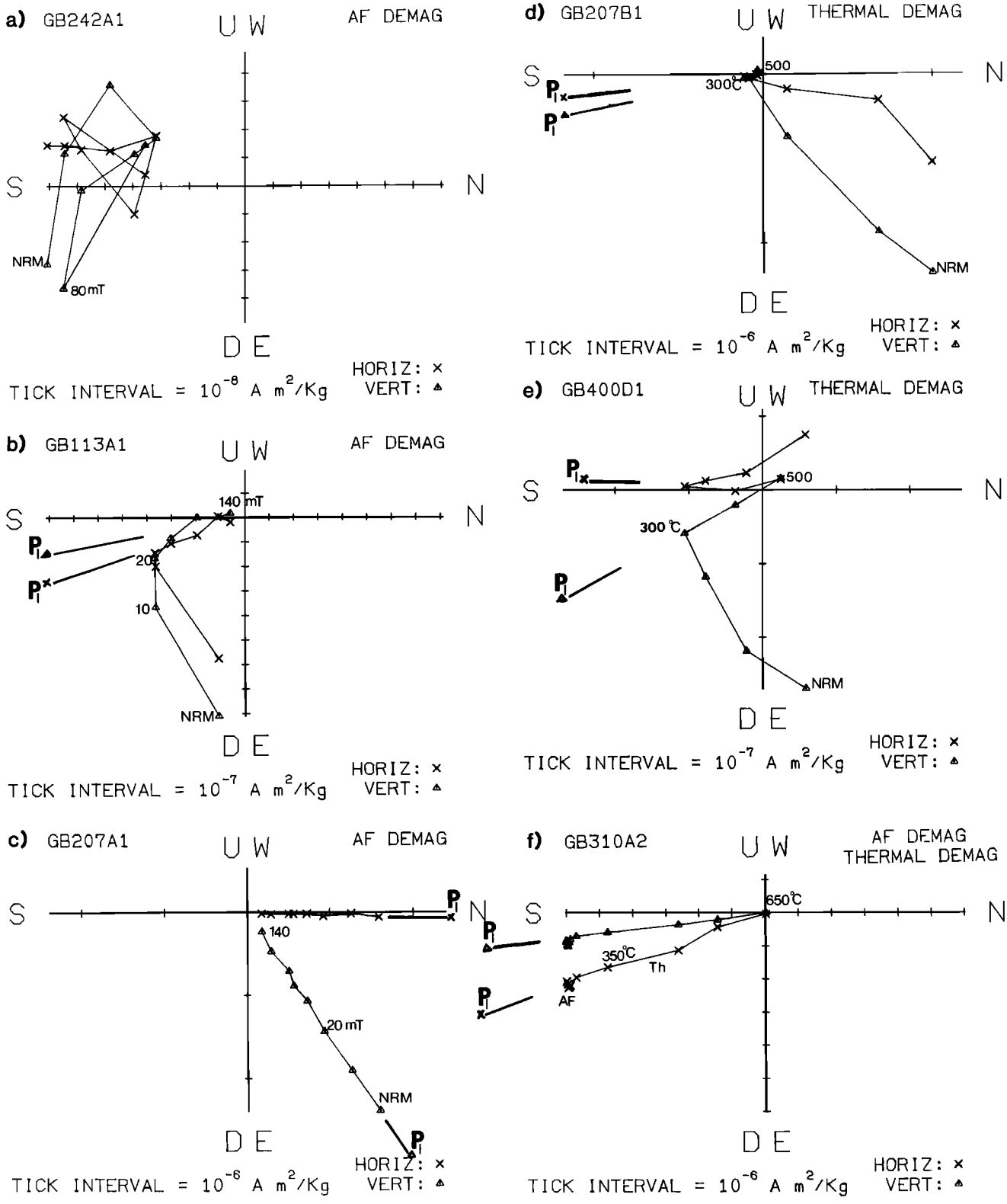


Fig. 3. Zijderveld vector demagnetization diagrams for selected gray limestones (a – e) and a red Taggard shale (f).

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particles too disseminated for identification under 1350x magnification.

Data Analysis

The classic multivariate technique of principal component analysis was applied to calculate the best least-squares fit of field direction [Schomaker et al., 1959; Blow, 1960; Kirschvink, 1980]. It has been applied at several different levels of the analysis. For individual specimens, it is used to determine the best-fitting straight-line segment in the Zijderveld demagnetization trace whose direction is taken as the specimen direction of characteristic magnetization. It may also be used as an alternative to Fisher [1953] statistics in determining the mean of a group of specimen, sample, or site directions. Principal component analysis is simply a linear transformation of the orthogonal coordinate axes to a new orthogonal reference frame that corresponds to the geometry of the data set. Discussion below is largely condensed from Kirschvink [1980].

Through the use of Lagrange multipliers, the principal axes parallel the eigenvectors of the matrix of sums of squares and products:

$$H = \begin{bmatrix} (x_i-\bar{x})^2 & (x_i-\bar{x})(y_i-\bar{y}) & (x_i-\bar{x})(z_i-\bar{z}) \\ (x_i-\bar{x})(y_i-\bar{y}) & (y_i-\bar{y})^2 & (y_i-\bar{y})(z_i-\bar{z}) \\ (x_i-\bar{x})(z_i-\bar{z}) & (y_i-\bar{y})(z_i-\bar{z}) & (z_i-\bar{z})^2 \end{bmatrix}$$

where $P_i = [x_i, y_i, z_i]$, i = 1 to N are the N data points in the old coordinate system and $U = [\bar{x}, \bar{y}, \bar{z}]$ is the position of the origin for the new coordinate system.

Eigenvectors of the matrix point along the axes of the new coordinate system and corresponding eigenvalues (λ_{max}, λ_{int}, λ_{min}) are the variance (σ^2) of the data along each principal axis. The relative magnitudes of the three eigenvalues can group a set of points along the demagnetization path into 3 categories, which are:

Line: $\lambda_{max} >> \lambda_{int} >= \lambda_{min} >= 0$

plane: $\lambda_{max} >= \lambda_{int} >> \lambda_{min} >= 0$

curve: $\lambda_{max} >= \lambda_{int} >= \lambda_{min} >> 0$

For example, if the points all lie precisely along a line, all of the variance is parallel to that line. The eigenvector associated with the maximum eigenvalue represents this principal component direction. The mean declination and inclination can thus be calculated from it. Similarly, the minimum eigenvector can be used to get the direction of a pole to a fitted plane for

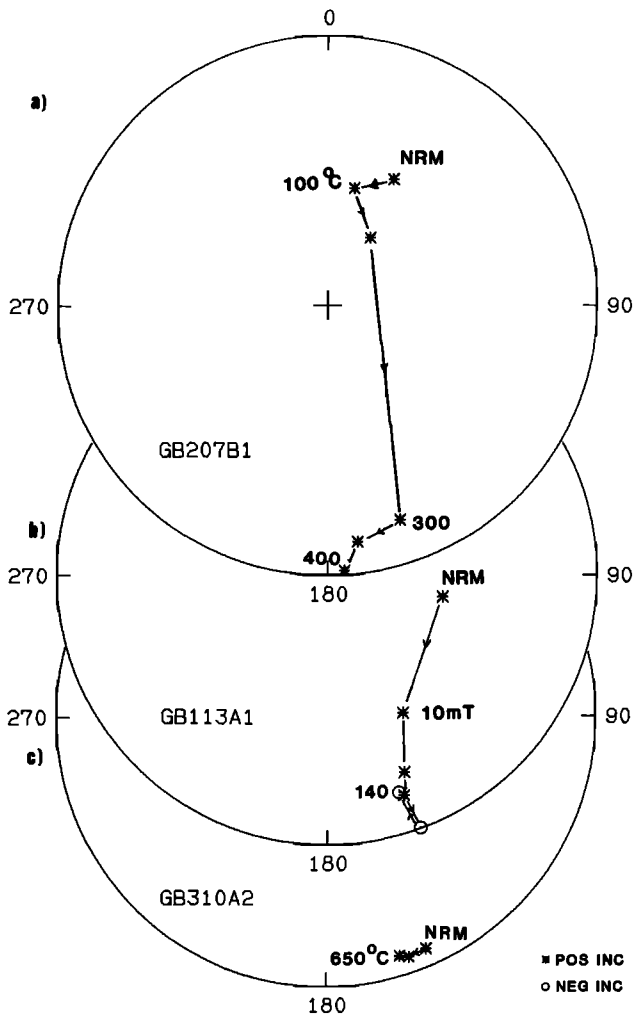


Fig. 4. Equal area projections for a) thermal, b) AF demagnetization of gray limestones, and c) thermal demagnetization of red sediments. All show field directions at the sampling site.

data which is dispersed in two dimensions, such as might be the case in a two-component demagnetization trace.

Kirschvink defines a useful quantity, the maximum angular deviation (MAD), which may be used as a rough basis for statistical inference. It is defined as the angle made to the fitted eigenvector (λ_{max} for a line, λ_{min} for a plane) by another vector which is the vector sum of all three eigenvectors whose lengths are given by their respective eigenvalues. For the case of the pole to a plane, the MAD is taken as 90° minus the angle just defined. The MAD is not analogous to α_{95} in Fisher statistics, since it is essentially independent of the number of data points being fitted. As a result it should be regarded more as an indicator of the noise level in the data set rather than as a measure of the

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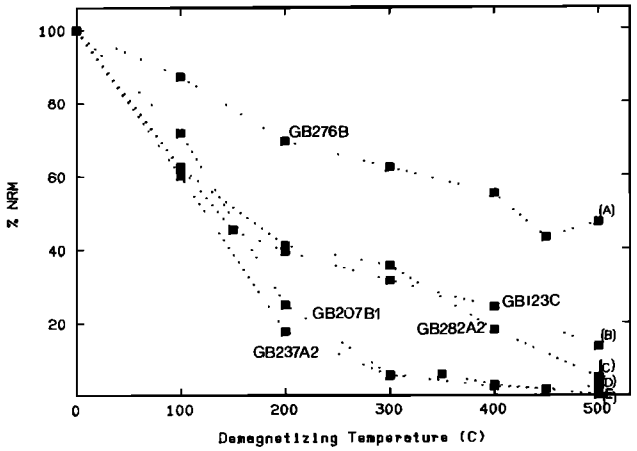


Fig. 5. Thermal decay of intensities of specimens from gray sediments normalized to unity at room temperature.

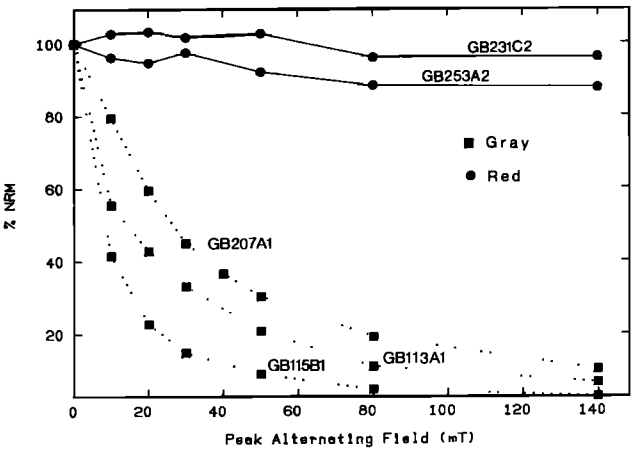


Fig. 6. Alternating field decay of intensities of specimens from gray and red sediments normalized to unity.

precision of the fit. The MAD is obtained from:

Line:

$$MAD = \tan^{-1}[(\lambda_{int}+\lambda_{min})/\lambda_{max}]^{1/2}$$

Plane:

$$MAD = \tan^{-1}[(\lambda_{min}/\lambda_{int})+(\lambda_{min}/\lambda_{max})]^{1/2}$$

Therefore, the MAD can serve as a paleomagnetic precision index which is purely geometric and bears little or no relationship to the demagnetization procedures used or to the number of demagnetizing steps. Small values indicate the collinearity or coplanarity of the data and may be used in a numerical data search. Large values can be induced by measurement error or the magnitude of superimposed magnetic components. The eigenvalues thus provide a valid basis for accepting or rejecting data from a particular specimen.

Since the origin of a Zijderveld diagram is an exact theoretical point on all demagnetization paths, and completely demagnetized specimens should have no remanence, the fitted demagnetization line was constrained to pass through the origin. Such an "anchored" demagnetization line is the least-squares generalization of a stable end-point direction, while an "anchored" plane is the similar generalization of a remagnetization circle.

Besides serving as a good method to estimate the directions of lines or planes of best fit along a demagnetization path, principal component analysis also functions well in deriving the mean direction of a paleomagnetic data set. For this purpose, the eigenvector corresponding to λ_{max} for a matrix which is anchored to the origin serves as the mean direction for the data set. Since the method is axial rather than directional in nature, mixtures of normal and reversed directions may be treated very conveniently.

The same matrix may be used to determine whether or not a data set is circularly distributed about the mean direction - a requisite for a Fisherian distribution. The ratio

$$a/b = (\lambda_{int}/\lambda_{min})^{1/2}$$

may be taken as the ratio of the major and minor semiaxes of an ellipse drawn about the fitted direction representing a sort of two-dimensional deviation about the mean. Thus, the extent to which the ratio a/b deviates from unity can be taken as a measure of the non-circularity of the data set about the mean. A circular and random data set, as is the widely used presumption for applying Fisher statistics, should have a value

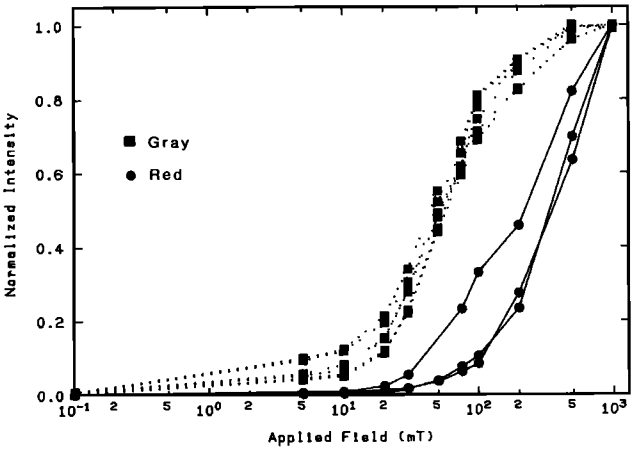


Fig. 7. Acquisition of isothermal remanence for selected gray and red sediments.

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close to one. The larger the value, the more elliptical the data set.

To determine if the cleaned results still contain contaminating components, a test of circularity should be performed before the proper statistics method is chosen. If the magnetic vectors for a set of specimens have a varying mixture of two or more superimposed magnetic components, the directional distribution of such specimens frequently may be non-Fisherian, in which case the application of Fisher statistics may turn out to be misleading.

To show the extent to which other published data may be contaminated by undetected components, two VGPs resulting from previous work were reviewed [Payne et al., 1981]. In Figures 9a and 9b field directions have been rotated such that their Fisher mean is centered on the diagram. The dashed circle is of arbitary radius and is for reference only. For the first example, the Buffalo siltstone with an ellipse axis ratio a/b = 2.9 , the distribution is non-Fisherian (elliptical) and the orientation of the major axis of the ellipse (short lines on the circle) is closely aligned with the great circle to the present axial dipole field direction (arrow) which is the main overprinting component. In the second example, the Brush Creek limestone, contamination is less as shown by a/b = 1.6, though alignment of the major axis of the ellipse with the axial field and slight asymmetry about the mean show that the present field is still a noticeable contaminant.

Ellipticity in the field direction data need not always imply contamination by a secondary component. For example, very high quality recorders of the magnetic field would be expected to show an elliptical distribution of field directions if the original set of virtual geomagnetic poles (VGPs) from which the field directions are derived were circularly distributed. In such a case the long axis of the ellipse would align with a great circle connecting the sampling site and the mean VGP. In many situations the alignment of the ellipse may make it possible to distinguish between this effect and contamination from a known secondary component such as the recent field.

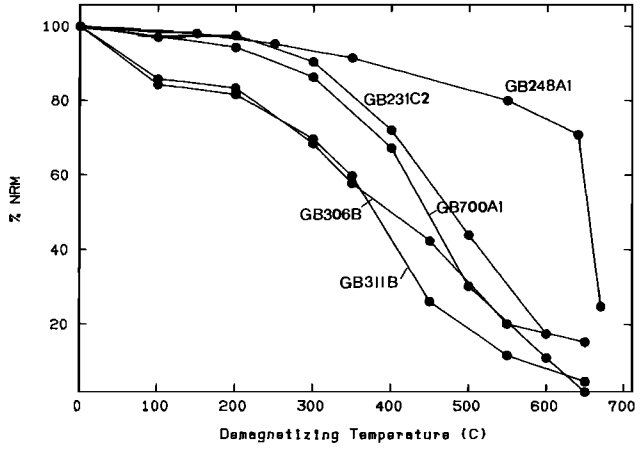


Fig. 8. Thermal decay of intensities of specimens from red sediments normalized to unity at room temperature.

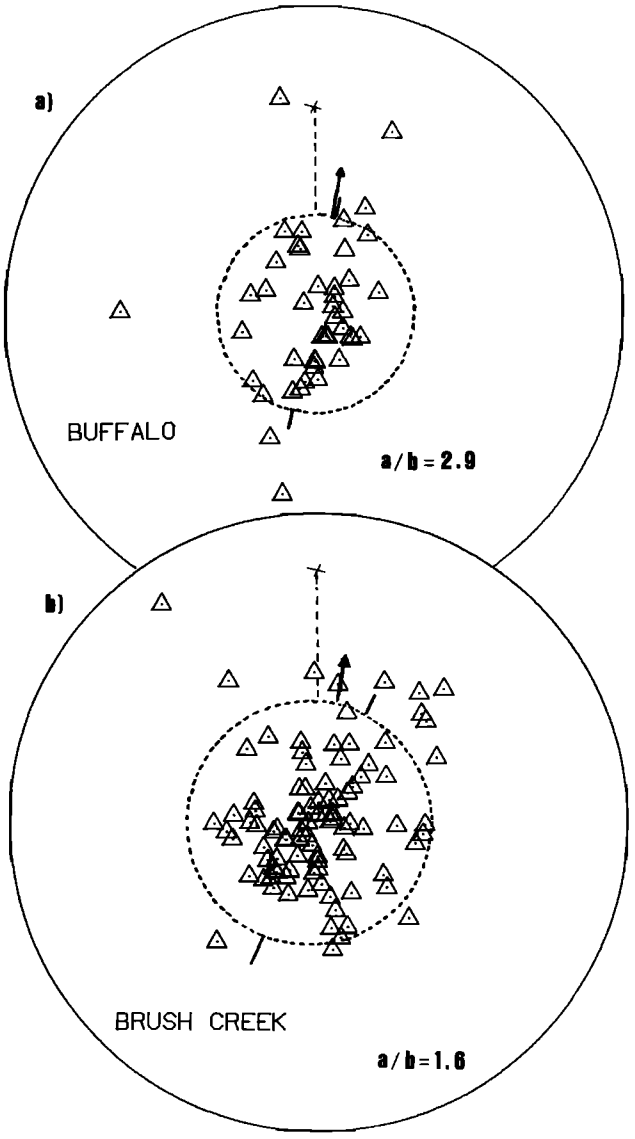


Fig. 9. Circularity diagrams for a) Buffalo siltstone and b) Brush Creek limestone [Payne et al., 1981]. Field directions are shown in equal area projection rotated such that the mean direction is centered on the diagram. Dashed circle is for reference only. Short lines on the circle denote direction of major axis of the elliptical distributions. Arrow denotes the direction to the axial dipole field. The vertical dashed line shows direction to the VGP.

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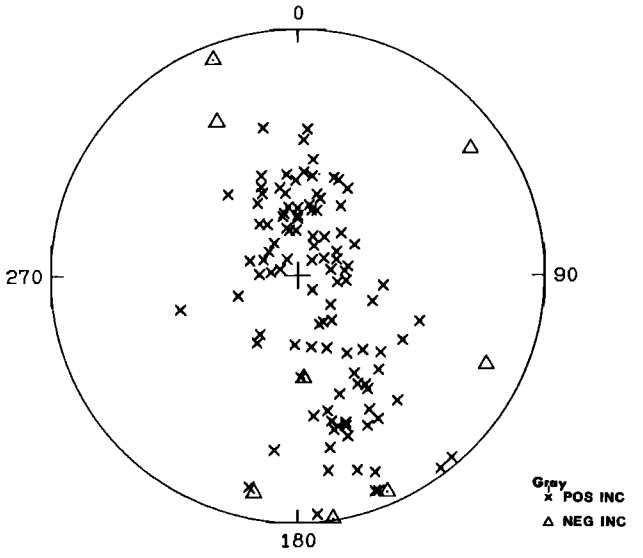


Fig. 10. Sample means after AF demagnetization for gray sediments.

Archeomagnetic fired-clay samples, for example, might well show this effect, since they often are very high-quality recorders.

In the case of many sediments, however, the main source of dispersion is random noise in the recording process, which should result in a circular distribution of field directions.

The effect of contamination by a secondary component is to shift the mean of the data set somewhat from the "true" characteristic direction toward that of the contaminating overprint. In this connection it is perhaps of interest to note that the Brush Creek pole reported by Payne et al. (with a/b = 1.6) agreed well with other Carboniferous poles, but the Buffalo pole (with a/b=2.9) did not. The latter is, in fact, displaced slightly in the direction of the present field (taking into account the fact that the polarity of the field was reversed).

If a data set is non-circular it should be treated instead with Bingham statistics [Bingham, 1964; Onstott, 1980]. Unfortunately, the application of Bingham statistics is computationally horrendous and reliable computer programs for the purpose are still under development [Onstott, personal communication]. Until such a program is generally available, principal component analysis offers at least a simple test for circularity of data sets.

Results and Discussion

Zijderveld plots for individual specimens were examined and linear demagnetization trajectories defined by three or more points were determined for the characteristic directions. The near orthogonality of the recent field and the southeasterly field vectors allowed us readily to distinguish one from the other. Principal component analysis was applied to this demagnetization path with the line constrained to pass through the origin, and the best least-squares fit was calculated as designated in Figures 3b to 3f by P_1. For each specimen, the criterion for goodness of fit was based on the MAD. Using a somewhat artibrary number, fitted directions with the MAD greater than 25° were discarded in the later calculation of mean sample directions.

Fisher statistics were then used for calculating sample means since usually less than five fitted directions were encountered in deriving sample means after selection. In the case of so few data points, the slight difference between the two methods would not be meaningful.

Fitted sample mean directions from all sites not corrected for bedding tilt are shown in Figure 10 for AF cleaning and Figure 11 for thermal cleaning. Since two discrete magnetic components are simultaneously removed in different ratios, wide variations in relative intensity and stability would produce a streaking of directions along a great circle passing through the characteristic direction and the recent field [e.g. Collinson and Runcorn, 1960; Robertson, 1966]. Thermal cleaning proved to best differentiate between these alternatives since the secondary component is thermally soft.

In addition to the strong clustering at the southeasterly edge of the net and the instability streak from the southeastern edge to the recent field, six samples were found to have a declination of 340°, with shallow inclinations, and oriented antipodally to the southeasterly clustering. They therefore represent a magnetization of normal polarity with respect to

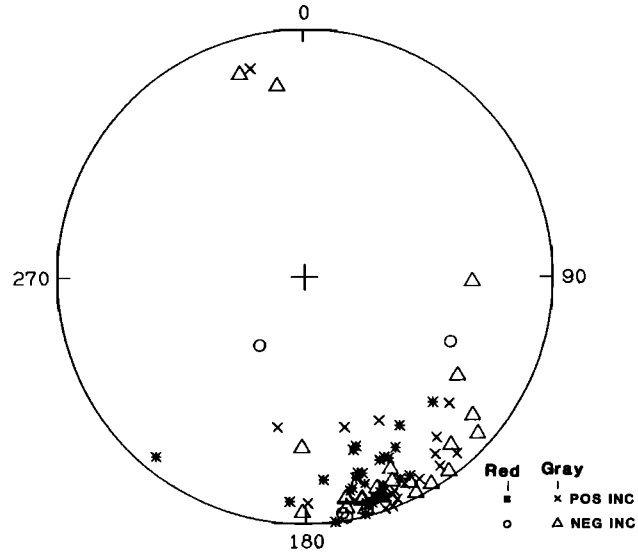


Fig. 11. Sample means after thermal demagnetization for gray and red sediments.

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the main clustering which is classified as being reversely magnetized.

Principal component line-fitting to all stable end-point normal recorders was first used. Following the selection criterion described earlier, four samples (Figures 10 and 11) gave good normal directions while the other two were discarded for larger angles of deviation. Among the four, one is the sample mean direction from four specimens after AF demagnetization, two are the directions from samples which yielded only one specimen for thermal demagnetization, and the last one is from one specimen for both AF and thermal processes.

All the normal recorders are gray sediments collected from outcrops with gentle bedding tilts in northern and central areas. Two samples came from each of the following sites: Huttonsville, Interstate-64, and Johnson Quarry. Correction for bedding did not have any significant effect on the magnetization directions. Normal polarity samples from Interstate-64 and Johnson Quarry sites could not be placed in their stratigraphic positions within the Greenbrier with certainty for lack of good marker zones. However, those two from Huttonsville (Site 3) were from the interbedded limestone layer between the Taggard red shales which indicates that more than one normal event may have been recorded. Subsequent work in progress has located a thin normal zone within the Taggard limestone layer at site 2 as well. McElhinny [1973, p. 127] shows two Mississippian normals from the U.S.S.R.

The magnetic appearance of these normal records varied noticeably, overprinted to a greater or lesser extent by the present field in both high and low coercivities and blocking temperatures, but fell within the range of behavior characteristic of the reversed recorders.

Though few in number, these normal samples yielded field directions very closely antipodal to the mean reversed directions. Their presence, especially among the thermally demagnetized samples (Figure 11) is a moderately persuasive argument that at least some of the gray sediments have retained their record of the Mississippian magnetic field. This is particularly true since normal events were so rare in the subsequent Kiaman reversed interval.

As thermal demagnetization was clearly more efficient in removing the contaminating overprint, it alone was used in deriving the final mean field direction for all samples. Again, only those sample means whose MAD (maximum angular deviation) was smaller than 25° were used in the calculation of the site means.

Using this criterion, 56 of the 115 thermally treated gray samples and one of the 32 red samples were eliminated. No useful data were obtained from the samples of site 4. In most cases the eliminated samples yielded only curved demagnetization trajectories which showed an excessive present field overprint. Remagnetization circle analysis appeared unprofitable since the demagnetization tracks for each sample were so similar that there was little convergence.

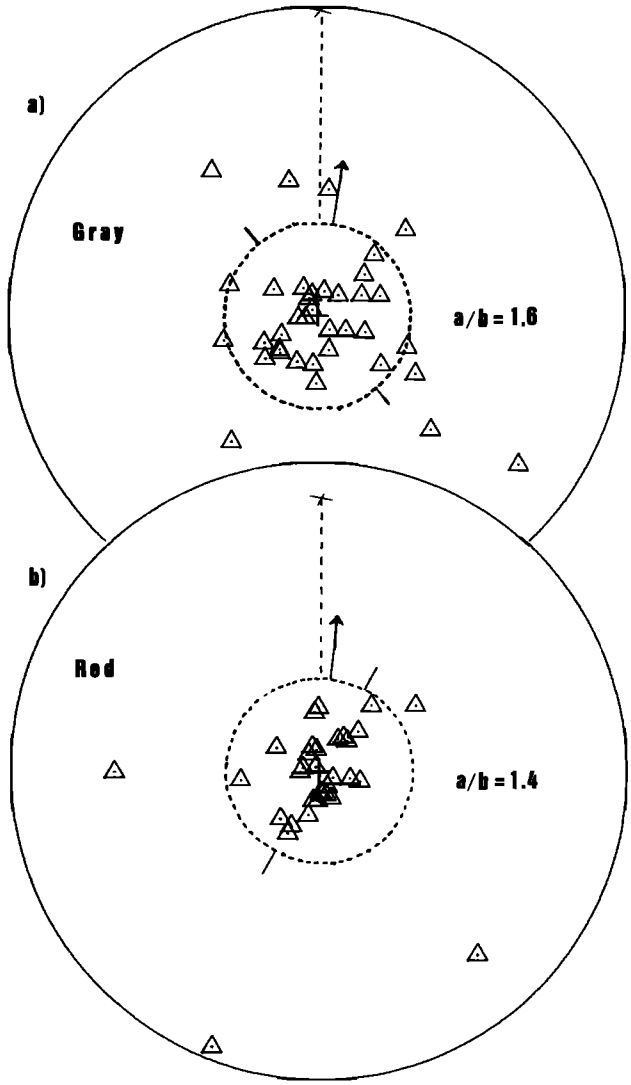


Fig. 12. Circularity diagrams for thermally cleaned a) gray and b) red sediments from the Greenbrier Group. See Figure 9 for explanation.

The site means calculated by Fisher statistics are listed in the Table. With the exception of sites 9, 10, and 11, the effect of a structural correction on the data shown in Figures 10 and 11 is quite small, due partly to the moderate dips and partly to the rather small angle between the mean magnetization direction and the axis of folding. Sites 9-11, however, are steeply dipping and in fact are overturned (see Appendix). Since uncorrected data from these sites are included in Figures 10 and 11, it is clear that the samples from at least those sites have been remagnetized.

Figure 12 shows circularity diagrams for both gray and red facies, uncorrected for bedding

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TABLE 1. Site Means

Site	Location	N	Before Tilt Correction Decl	Incl	k	α95	After Tilt Correction Decl	Incl	k	α95
Gray Sediments										
1	38.92N 79.86W	4	161.1	2.5	17.7	22.5	163.0	13.3		
2	38.91N 79.71W	3	151.7	13.5	8.5	45.2	152.9	10.8		
3	38.66N 79.92W	4	158.0	20.6	24.5	19.0	156.5	12.0		
5	37.75N 80.48W	13	152.8	1.5	34.8	7.1	153.0	5.9		
6	37.87N 80.55W	7	163.7	-0.5	21.4	13.3	166.8	14.6		
7	37.74N 80.54W	5	151.2	4.0	14.9	20.5	152.5	12.1		
8	37.61N 80.57W	6	160.7	-10.4	132.1	5.9	160.7	-10.4		
9	37.32N 81.10W	5	158.7	-7.0	40.1	12.2	53.5	25.8		
10	37.37N 81.10W	9	174.2	26.0	22.5	11.1	6.4	14.8		
Means										
Sites 1-3, 5-10		9	159.0	5.6	33.6	9.0	144.8	16.1	2.8	37.9
Sites 1-3, 5-8		7	157.0	4.5	52.4	8.4	157.9*	8.4*	61.8	7.7
Red Sediments										
1	38.92N 79.86W	7	172.7	0.0	5.2	27.4	173.8	7.9		
2	38.91N 79.71W	6	161.8	13.3	19.8	15.5	162.8	9.7		
3	38.66N 79.92W	6	167.7	10.9	76.8	7.7	166.9	3.3		
7	37.74N 80.54W	4	161.8	7.1	25.1	18.7	163.7	13.3		
11	37.25N 81.21W	8	157.6	15.5	10.0	18.4	323.1	21.7		
Means										
Sites 1-3,7,11		5	164.4*	9.4*	93.3	8.0	173.7	17.2	2.3	66.3
Sites 1-3, 7		4	166.0	7.9	108.8	8.8	166.8	8.6	158.1	7.3

*Directions used for the calculation of VGPs plotted in Figure 15.

tilt. They display ellipticity ratios of 1.6 and 1.4, respectively, indicating nearly circular distributions.

Though the samples taken all together appear to show a circular distribution, the site means listed in Table 1 (either before or after tilt correction) do not. The circularity test described here is meaningful only for a fairly large number of data points, and for few points it tends to respond strongly to individual anomalous points, particularly those which lie farthest from the mean.

Age of Magnetization

After a period of extensional tectonics accompanying the main subsidence in Ordovician time, deformation of the Valley and Ridge Province by a period of compressional forces ensued in late Mississippian to early Permian time, during which the final group of clastic sediments were deposited and the whole section was deformed [Rodgers, 1972].

To determine the relative age of the stable paleomagnetic direction, the fold test [Graham, 1949] was applied.

Site mean directions of all thermally cleaned data derived by giving unit weight to each mean sample direction are plotted in Figures 13 and 14 for the same data after and before the correction for bedding tilt has been applied. There is an increase in dispersion after the bedding correction. To determine the increase in dispersion for significance, the test proposed by McElhinny [1964] was used. The ratio for the precision parameter k before and after folding is computed and compared with values tabulated by McElhinny. It shows that the test is negative where k_b/k_a is 12.0 for gray sediments from 7 sites, and is 40.6 for red sediments from 5 sites. The test is significant at the 99% confidence level.

In view of the earlier discussion, it seems fairly clear that McElhinny's use of Fisher statistics in the fold test is suspect, since at least one of the distributions (either before or after unfolding) is almost certainly non-circular. The test should properly involve Bingham statistics and should also determine if the increase or decrease in circularity of the distribution is statistically significant. Fortunately, in the present case, the fold test

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is clear-cut on visual inspection. The general problem remains, however, and is deserving of attention by statisticians.

While the results of the fold test for the gray sediments appear at first to be very clear (Figure 13), results for the red sediments are dependent on a single site (site 11 - Figure 14). As luck would have it, the beds are overturned in just such a way as to imply that the sediments at this site might have been normally magnetized, with a resultant null fold test (Figure 14a). This is not at all likely, however, since eight samples were taken at this site, providing thorough coverage of the red members of the Taggard Formation. Good coverage was also obtained at other sites, but none of those samples yielded normal directions. Hence we interpret the fold test in Figure 14 as convincingly negative, at least for site 11.

Though the other sites show dips of as much as 29°, the magnetization direction is too close to the structural axis to show any substantial effect on unfolding.

Conclusions

The results of this study appear at first glance to be contradictory. The existence of normal as well as reversed poles for the gray sediments indicates that remagnetization has not

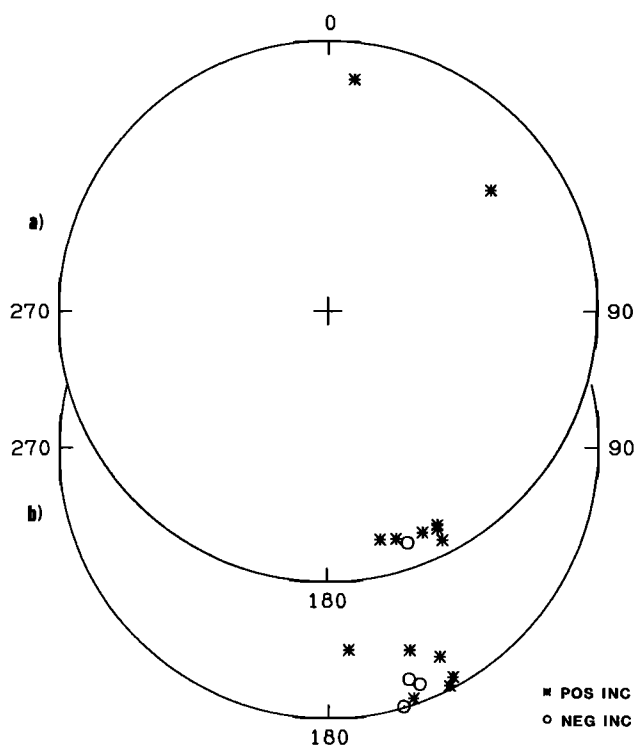


Fig. 13. Site mean directions for thermally cleaned gray limestones a) after and b) before correction for the bedding tilt.

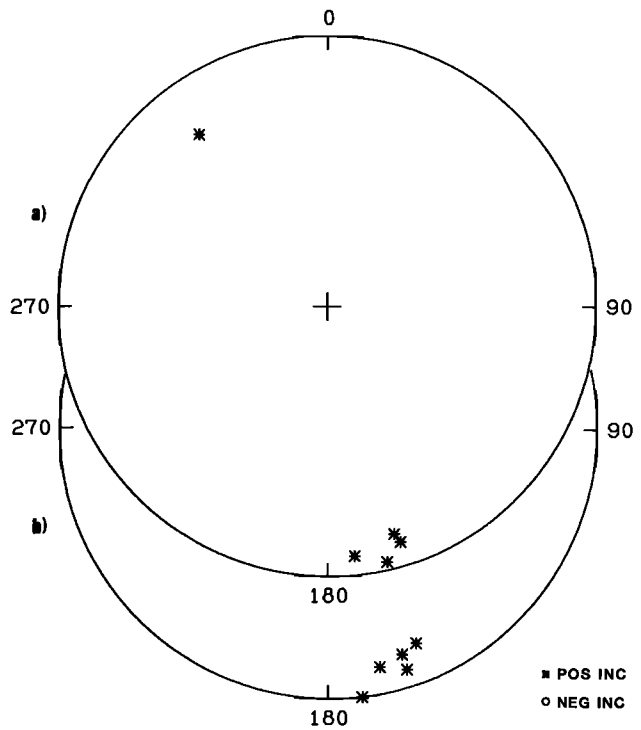


Fig. 14. Site mean directions for thermally cleaned red sediments a) after and b) before correction for the bedding tilt.

taken place, while the negative fold test for both gray and red sediments shows rather clearly that it has.

A possible resolution to this problem lies in the fact that these two conflicting results come from mutually exclusive sites. The normal polarities come from sites 3, 5, and 6, while the significant fold test results come from sites 9, 10, and 11.

Of interest in this regard is the conodont color alteration index (CAI) variation for Upper Devonian through Mississippian rocks in the study area [Figures 17 and 18 in Epstein et al., 1977]. Sites 3, 5, and 6 fall in an area where the CAI is in the range 1.5 to 2, while sites 9, 10, and 11 fall in an area where the CAI ranges from 2.5 to 3.5. According to Epstein et al., this represents a temperature range of 50°-140°C for sites 3, 5, and 6, and approximately 85°-250°C for sites 9, 10, and 11.

Temperatures of around 200°C for extended periods of time are likely to reset the gray sediments, at least partially (see Figure 5), but are less likely to reset the red beds (Figure 8). There is also the real possibility that the hematite of the red beds is purely diagenetic, and represents a magnetization acquired long after deposition of the rock units [Larson and Walker, 1982]. The close correspondence between unfolded red sediment results from the different

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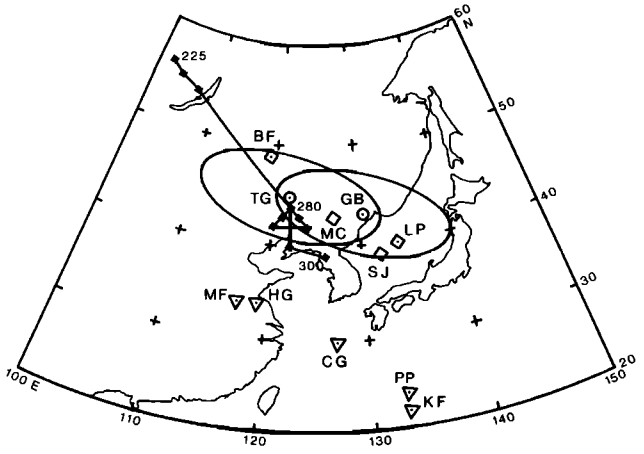


Fig. 15. Paleopoles from Mississippian units. Circles: (with 95% ovals of confidence): GB - Greenbrier limestones,sites 1-3, 5-8, structurally corrected, thermally cleaned (this study); TG - Taggard red sediments, all sites, structurally uncorrected, thermally cleaned [this study]. The Taggard is considered a Permian remagnetized pole. Diamonds (Cratonic North America): LP - Lilesville pluton, N. C. [Barton and Brown, 1983]; SJ - St. Joe Limestone, Ark. [Scott, 1979]; MC - Mauch Chunk Formation, Pa. [Knowles and Opdyke, 1968]; BF - Barnett Formation, Tex. [Kent and Opdyke, 1979]. Triangles (Acadian): KF - Kennebecasis Formation, N. B. [Black, 1964]; PP - Pre-Pictou Formation, N. B. (Black, 1964); CG - Codroy Formation, Nfld. (Black, 1964); HG - Hopewell Grioup, N. B. (Roy and Park, 1969); MF - Maringouin Formation, N. B. (Roy and Robertson, 1968). Points connected by line: 225, 230, 240, 250, 260, 270, 275, 280, 290, and 300 m.y. averaged Permian and Late Carboniferous poles [Irving, 1978].

sites in this study argues in favor of the latter interpretation.

The gray sediments may give another story, however. Those from sites 9 and 10 are clearly reset, but the others may not be, as evidenced by the presence of normal polarity samples.

If the three southeasternmost sites with the highest CAI indexes are omitted, the remaining sites (1-3, 5-8) for the gray sediments yield mean directions of I = 8.4°, D = 157.9° corrected for tilt. The tilt correction has little effect on precision index (k) values and circularity ratios (a/b = 1.6 before tilt correction and a/b = 1.5 after correction). The corrected field direction yields a reversed polarity VGP at S 43.2° W 49.5° with dp = 3.9° and dM = 7.8° which is plotted as a normal pole (point GB) along with its oval of 95% confidence in Figure 15.

There is a fair probability that this pole for the gray sediments has paleomagnetic significance, since it is based on sites with a low CAI index and includes at least three sites in which mixed polarities are observed. On the other hand, there is no fold test for these sites and the uncorrected gray site means for the heavily folded sites 9 and 10 are not very different.

For these reasons, we would regard this pole for the Greenbrier Group gray sediments as poor to fair in reliability. We should note, however, that if the overturned southeastern outcrops had not existed, then our results probably would have appeared favorable in comparison with most other published Mississippian cratonic poles (Figure 15), only one of which had a fold test. Indeed, the pole from this study is in good agreement with that pole, which is from Knowles and Opdyke's [1968] study of the folded Mauch Chunk Formation of central Pennsylvania.

At the present time, the best support for the reliability of Mississippian cratonic poles is the great diversity of rock types and locations from which they are derived and their agreement with one another.

Point TG (and its oval of confidence) in Figure 15 is derived from the uncorrected Taggard red bed mean derived from all the sites where it appears (see Table). This pole should represent a post-folding remagnetization in Pennsylvanian or later time. In fact, it falls nearly on Irving's [1979] averaged 250 - 280 m.y. poles and is in excellent agreement with Helsley's [1965] Permian pole for the nearby Dunkard Series. Though it is statistically indistinguishable from the gray sediment pole, and from all the other cratonic Mississippian poles as well, it nonetheless suggests an Early- to mid-Permian remagnetization time for the Taggard red beds. The close coincidence between the Carboniferous and Permian poles is clearly a source of considerable difficulty in interpretation of Carboniferous results. The Taggard pole will not be considered further here.

The present study, with its difficulties, seems typical of the problems encountered in cratonic North American Carboniferous paleomagnetism and suggests that the only way in which a reliable Paleozoic cratonic pole path can be determined is to compile a sufficient quantity of data from as many different locations and lithofacies as possible in hopes of separating signal from noise.

In spite of these difficulties, the scant data already available has led to extremely interesting interpretations. Figure 15 plots Mississippian cratonic poles as diamonds and Acadian poles of similar age as triangles. The two sets of five poles each are clearly set apart from one another. This, along with similar Devonian data, has led to the model of the Acadian displaced terrane advanced by Kent and Opdyke [1978] and discussed by Irving [1979] and Van der Voo [1982] among others. Van der Voo,

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French, and French [1979] show a relative displacement of 18° in paleolatitude between Acadia and cratonic North America for Late Devonian time. The sets of poles in Figure 15 are displaced from one another by 14°, or, if the Barnett pole is not considered cratonic [Irving, 1979], the displacement is 13°. Thus, some 4° or 5° of latitudinal motion (about 500 km north-south displacement) appears to have occurred between Late Devonian and Mid-Mississippian times, leaving some 1500 km of north-south displacement remaining before docking occurred in the Pennsylvanian.

These figures indicate that the north-south component of Acadian motion relative to North America increased during Mississippian and earliest Pennsylvanian time from approximately 2.5 cm/yr to 4.3 cm/yr. Strike-slip motion of the sort required by Van der Voo [1982] would add an east-west component that would increase these figures by about 40%. The lower rate may have prevailed throughout much of Mississippian time as evidenced by the persistent gap between the two groups of poles in Figure 15.

It is hard to ignore the linear and nearly parallel trends of the cratonic and Acadian sets of poles. The extreme ellipticities of the two sets (a/b = 36 and 82, respectively) seem unlikely from pure happenstance. The orientation of the trend is wrong for effects due to contamination by the present field, but is approximately correct for distortion due to the field direction to VGP conversion. The ellipticities seem much too high for this explanation, however, since the conversion can only produce a maximum ellipticity of 2.

A possible cause is a greater or lesser extent of Permian overprint, though the Maringouin and Hopewell poles, which were subjected to thorough and careful cleaning techniques, are more displaced toward the Permian than the other Acadian poles which received relatively rudimentary demagnetization.

It is tempting to regard these tracks as real, and to determine a genuine location for Acadia relative to North America by mapping the one track onto the other. In fact, this yields an Eulerian rotation (14.3° counter-clockwise about N 0.8° E 23.6°) which places Acadia only a few degrees to the southeast of Van der Voo's [1982] Late Devonian position.

This approach is probably not tenable, however, for several reasons: 1) the relative motion of Acadia with respect to North America distorts the orientation of the tracks with respect to one another; 2) the time sense is unclear, with a vague indication of time proceeding along the tracks from the southeast to northwest on the basis of the age of the sampled units, but in the opposite sense according to the longer-term track shown in Scott [1979]; and 3) the Devonian Acadian data do not seem to continue the track in any meaningful way [Kent and Opdyke, 1978].

At the present time we cannot conclude whether or not these tracks have any real significance.

APPENDIX

Sampling Sites

Site: 1
Name: Elkins Quarry
Lat,Long: N 38.92, W 79.86
Bedding: 26/17.5/W

The site is a limestone quarry (Elkins Limestone Company) located near Elkins along U. S. Route 219. Section is fairly complete, and includes the Taggard formation and upper Greenbrier Group.

Site: 2
Name: State-33
Lat,Long: N 38.91, W 79.71
Bedding: 312/6/W

The site is a road-side outcrop along U. S. Route 33 exposing the whole section of the Greenbrier Group, approximately 0.5 miles south of Bowden.

Site: 3
Name: Huttonsville
Lat,Long: N 38.66, W 79.92
Bedding: 223/10/S

In central Randolph County, the Greenbrier sequence is well exposed on the west slope of Cheat Moutain along U. S. Route 250, about 3.2 miles southeast of Huttonsville. Possible formations include Taggard, Pickaway, Union and Alderson.

Site: 4
Name: Edray
Lat,Long: N 38.28, W 80.12
Bedding: Horizontal

Greenbrier limestones outcrop just northwest of Edray along the west side of U. S. Highway No. 219 ascending Wolfpen Ridge in central Pocahontas County. Strata are nearly horizontal. Possible formations consist of Taggard, Pickaway, Union and Alderson.

Site: 5
Name: Interstate-64
Lat,Long: N 37.75, W 80.48
Bedding: 220/5/W

The locality consists of a series of road-cut outcrops along Interstate Route 64. Sample collecting starts near the Lewisburg interchange and extends westward nearly to the junction with U. S. Route 60. Limestones are the major rocks found.

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Site: 6
Name: Johnson Quarry
Lat,Long: N 37.87, W 80.55
Bedding: 22/29/W

The site is a limestone quarry (Johnson Limestone, Inc) along U. S. Route 60 about 12 miles west of Lewisburg, West Virginia. Section consists of limestone formations only, possibly Pickaway and Union.

Site: 7
Name: Fort Spring
Lat,Long: N 37.74, W 80.54
Bedding: 20/12/W

Section starts 1.2 miles east of Acme Quarry entrance at red Taggard exposure on west side of blacktop road. Only Taggard and Pickaway were found.

Site: 8
Name: Union
Lat,Long: N 37.61, W 80.57
Bedding: Horizontal

About 2.8 miles north of the town of Union in Union County, Greenbrier limestones are exposed along a dirt lane ascending Bickett Knob from the north. Possible formations are Pickaway, Union, Alderson.

Site: 9
Name: Abandoned Quarry
Lat,Long: N 37.32, W 81.10
Bedding: 272/50/S (Section overturned)

The site is an abandoned limestone quarry on the west side of State Route 12 near Ingleside. Limestones are the major lithofacies found.

Site: 10
Name: Mercer Quarry
Lat,Long: N 37.37, W 81.10
Bedding: 267.5/41/S (Section overturned)

The site is a limestone quarry (Mercer Crushed Stone, Inc.) located about 4 miles from U. S. Interstate 77 exit no. 5. Possible formations include Pickaway, Union, Alderson.

Site: 11
Name: Bluefield
Lat,Long: N 37.25, W 81.21
Bedding: 246.5/37.5/S (Section overturned)

Site located at ramp onto Interstate Route 77 near Bluefield (last exit near the boundary of West Virginia and Virginia). Taggard red shales were the only samples collected.

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